

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.34695225973635535, median 0.3138960542750081, std: 0.19619534773677447
Reprojection error (cam1): mean 0.34621132665482907, median 0.31437376237489567, std: 0.19587905985494652
Gyroscope error (imu0): mean 0.5424078725156954, median 0.4228497950155237, std: 0.5336541898168979
Accelerometer error (imu0): mean 0.38735413284545206, median 0.30219703048097973, std: 0.37193025981919964

Residuals

Reprojection error (cam0) [px]: mean 0.34695225973635535, median 0.3138960542750081, std: 0.19619534773677447
Reprojection error (cam1) [px]: mean 0.34621132665482907, median 0.31437376237489567, std: 0.19587905985494652
Gyroscope error (imu0) [rad/s]: mean 0.00748702155318649, median 0.005836724888151591, std: 0.0073661917969152875
Accelerometer error (imu0) [m/s^2]: mean 0.07525021474817892, median 0.058706980284185424, std: 0.07225386164630761

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.99999735 0.00225728 0.00045537 0.00551963]
[-0.00225656 0.99999621 -0.00157959 -0.00675094]
[-0.00045893 0.00157856 0.99999865 -0.02370046]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):
[[0.99999735 -0.00225656 -0.00045893 -0.00554572]
[0.00225728 0.99999621 0.00157856 0.00677587]
[0.00045537 -0.00157959 0.99999865 0.02368725]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)
0.003624639241073109

Transformation (cam1):

T_ci: (imu0 to cam1):
[[0.99999683 0.00216246 -0.00129355 -0.04463081]
[-0.00216405 0.9999969 -0.00123203 -0.00673284]
[0.00129088 0.00123482 0.9999984 -0.02363865]
[0. 0. 0. 1.]]

T_ic: (cam1 to imu0):
[[0.99999683 -0.00216405 0.00129088 0.04464661]
[0.00216246 0.9999969 0.00123482 0.00685852]
[-0.00129355 -0.00123203 0.9999984 0.02357259]
[0. 0. 0. 1.]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.003623994468905067

Baselines:

Baseline (cam0 to cam1):
[[0.99999847 -0.00009206 -0.00174907 -0.0501925]
[0.00009267 0.99999994 0.00034752 0.00002582]
[0.00174903 -0.00034769 0.99999841 0.00004977]
[0. 0. 0. 1.]]
baseline norm: 0.05019253234553865 [m]

Gravity vector in target coords: [m/s^2]
[-0.17645219 -9.80492043 -0.02868534]

Calibration configuration
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cam0

Camera model: pinhole
Focal length: [413.5111469032598, 414.20795715939494]
Principal point: [423.521867960405, 239.42372024086853]
Distortion model: radtan
Distortion coefficients: [-0.007659222164196881, -0.0005023682466075009, 0.0010448958953810197, -0.0004827913440815677]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [413.60804959761293, 414.3306050760599]
Principal point: [424.3561477013585, 239.32237311647756]
Distortion model: radtan
Distortion coefficients: [-0.007257102060135205, -0.0008461974580259748, 0.0011936590138093637, -6.837457881429248e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration

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IMU0:

Model: scale-misalignment
Update rate: 200

Accelerometer:

Noise density: 0.01373676763

Noise density (discrete): 0.1942672308551372

Random walk: 0.0005624771742904598

Gyroscope:

Noise density: 0.00097604109

Noise density (discrete): 0.013803305469114187

Random walk: 1.4185449668488343e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

[0. 1. 0. 0.]

[0. 0. 1. 0.]

[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

Gyroscope:

M:

[[0.99644643 0. 0.]

[0.00114061 0.98951477 0.]

[-0.00138482 0.00192741 1.00072278]]

A [(rad/s)/(m/s²)]:

[[0.00019587 -0.0005249 0.0003415]

[0.00057046 0.00052931 0.00012377]

[-0.0003832 -0.00139339 0.00011531]]

C_gyro_i:

[[0.99998587 0.0046873 -0.00250608]

[-0.0046829 0.99998749 0.00175787]

[0.00251429 -0.00174611 0.99999531]]

Accelerometer:

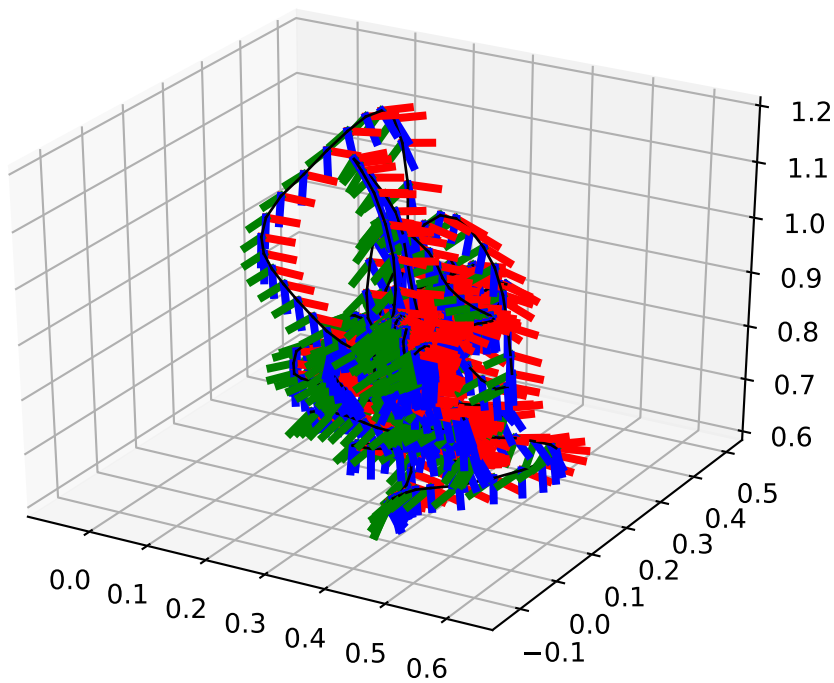
M:

[[1.0004857 0. 0.]

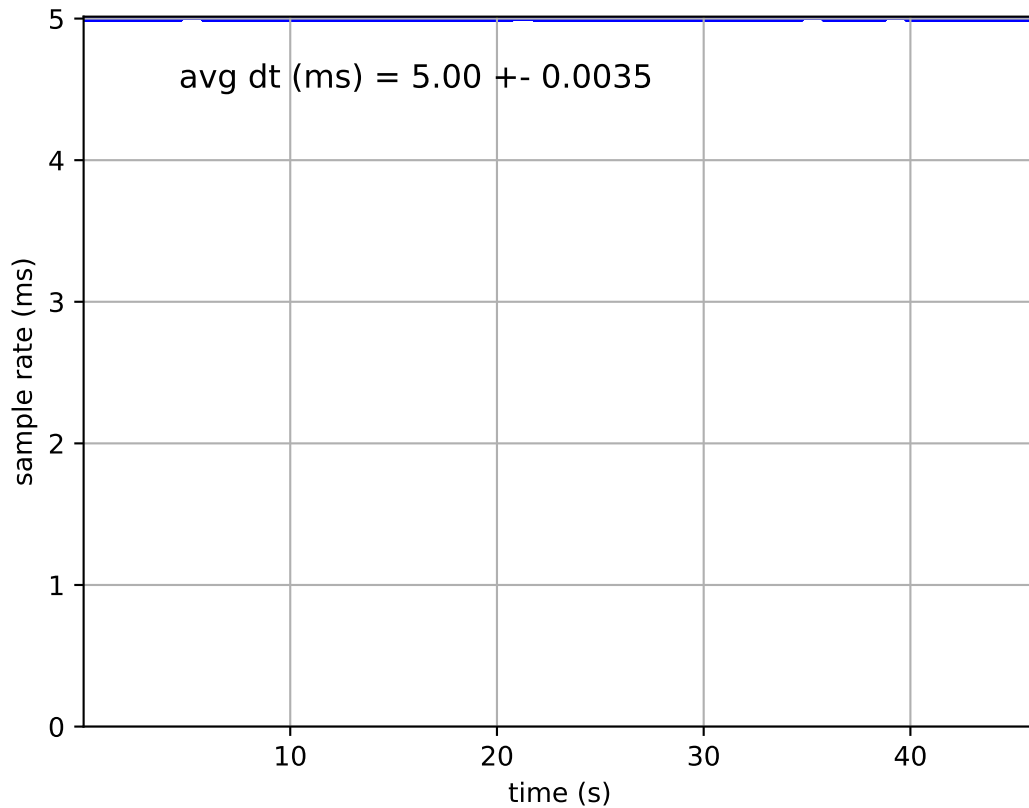
[-0.05387324 1.00223905 0.]

[-0.00001725 0.00371218 0.99968928]]

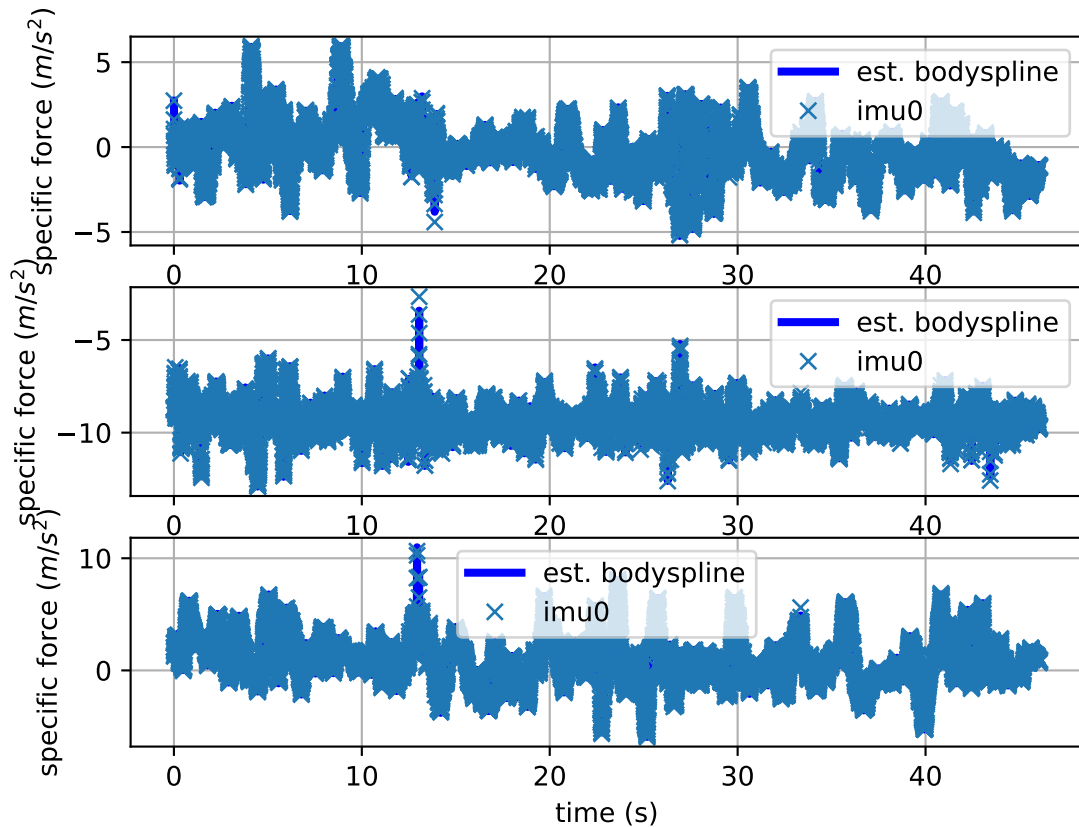
imu0: estimated poses



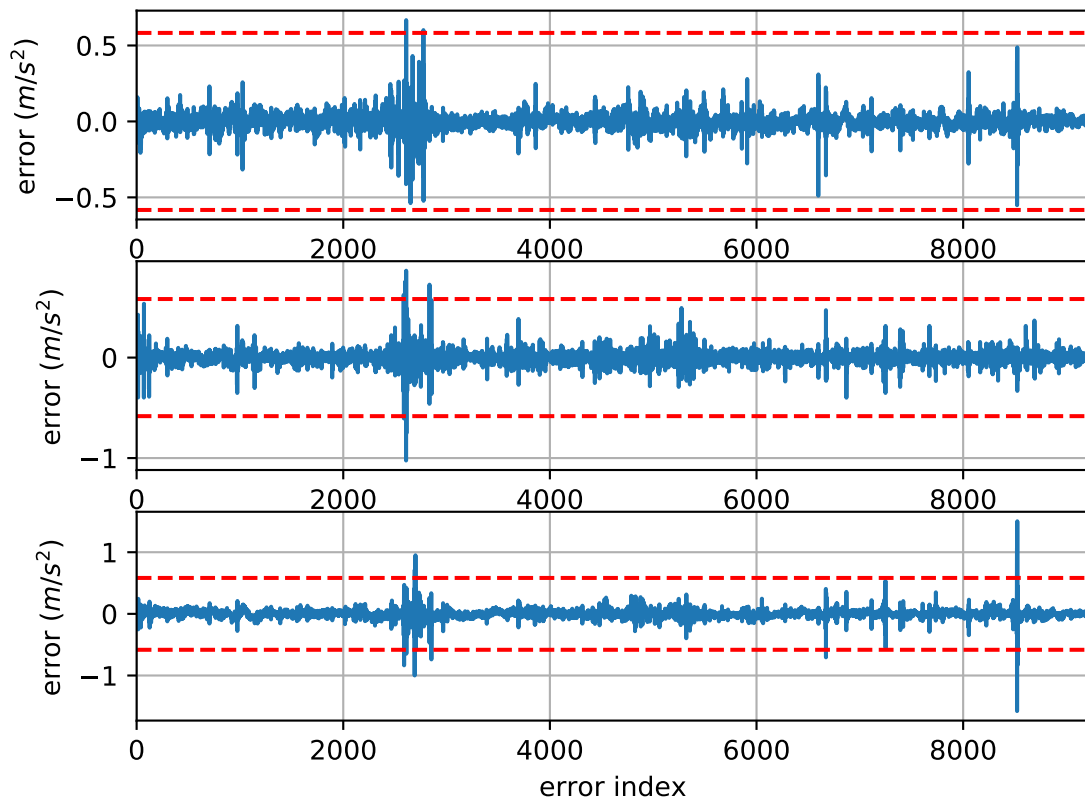
imu0: sample inertial rate



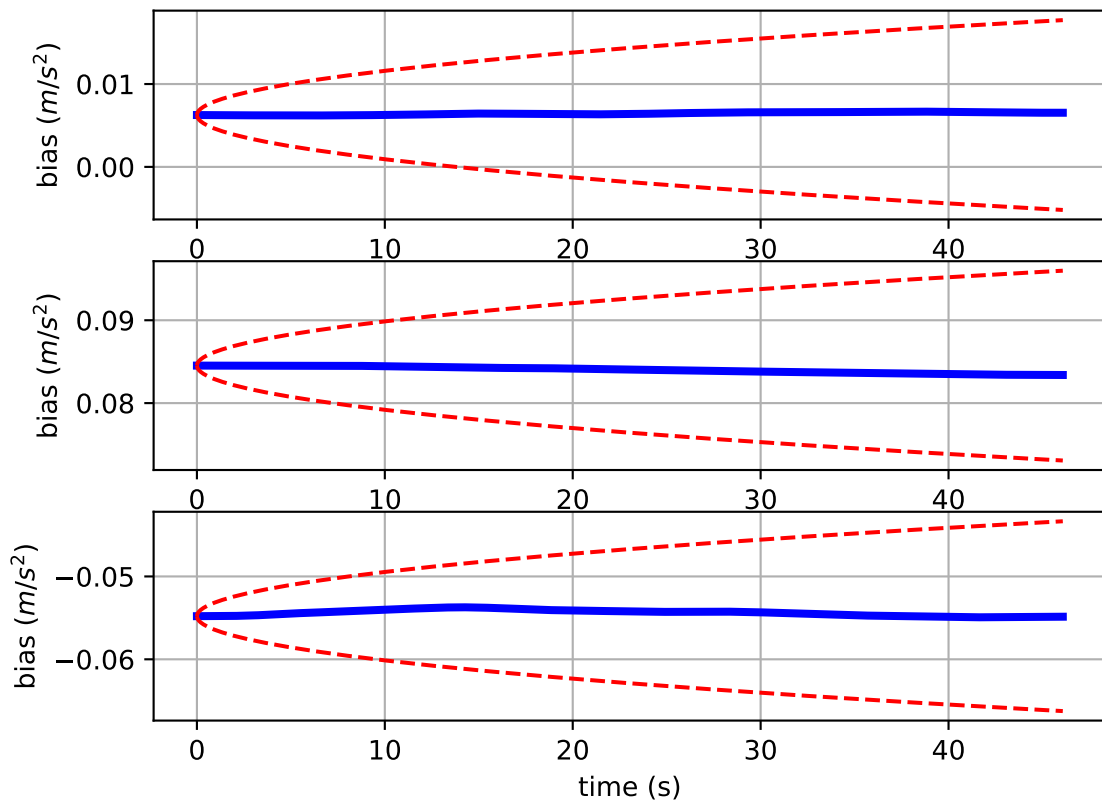
Comparison of predicted and measured specific force (imu0 frame)



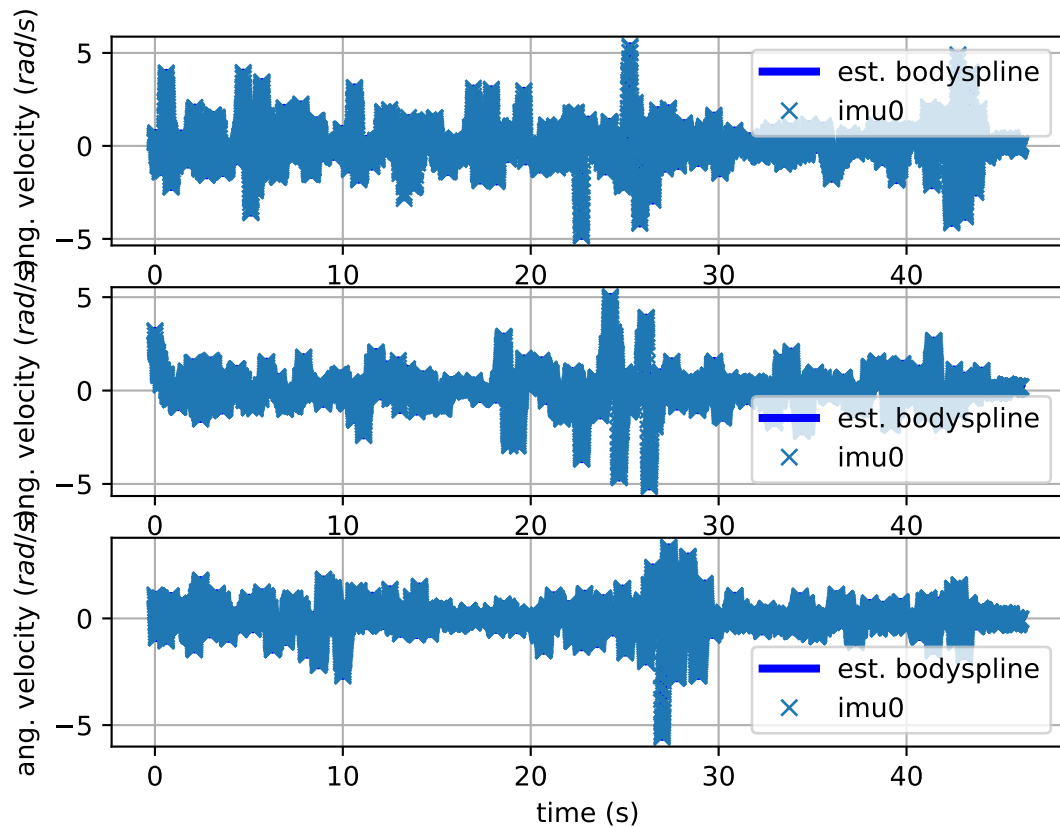
imu0: acceleration error



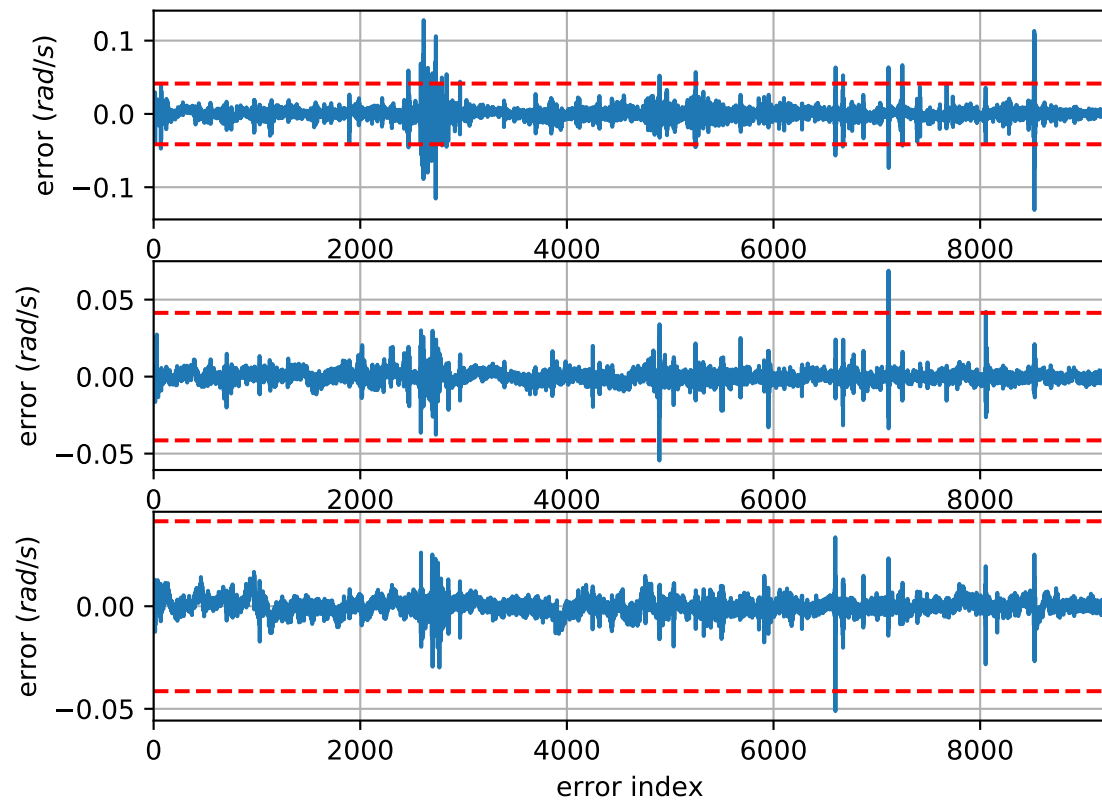
imu0: estimated accelerometer bias (imu frame)



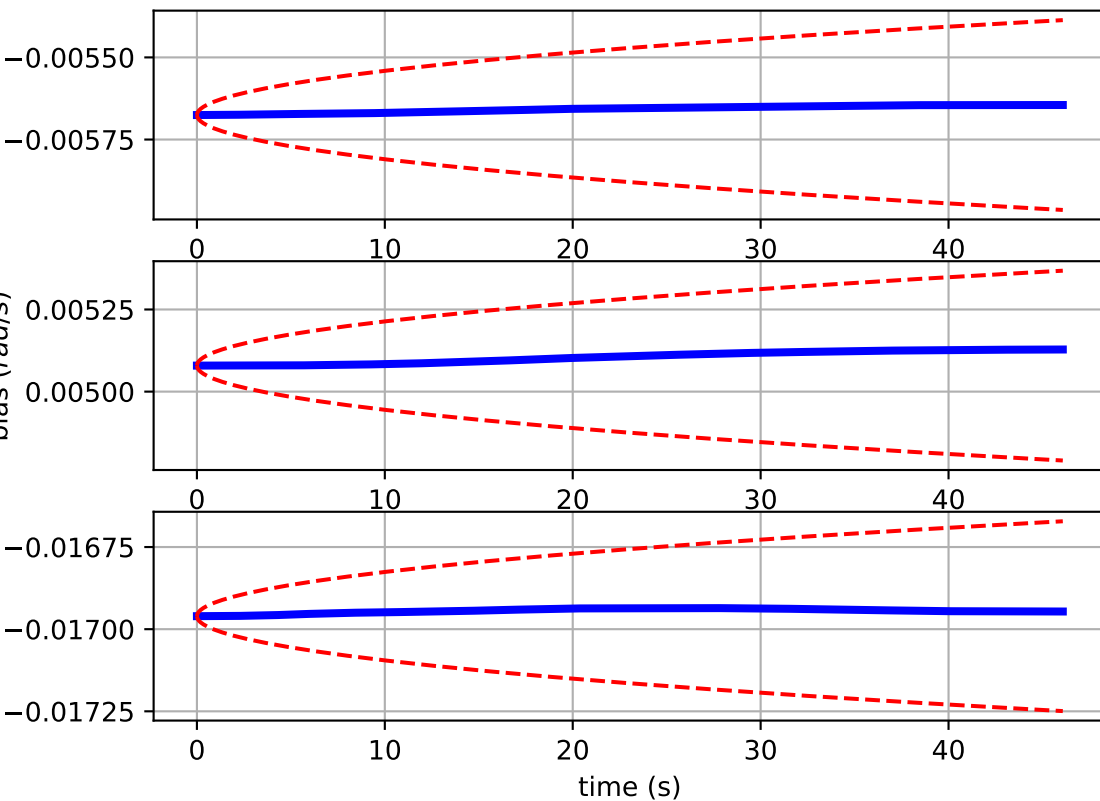
Comparison of predicted and measured angular velocities (body frame)



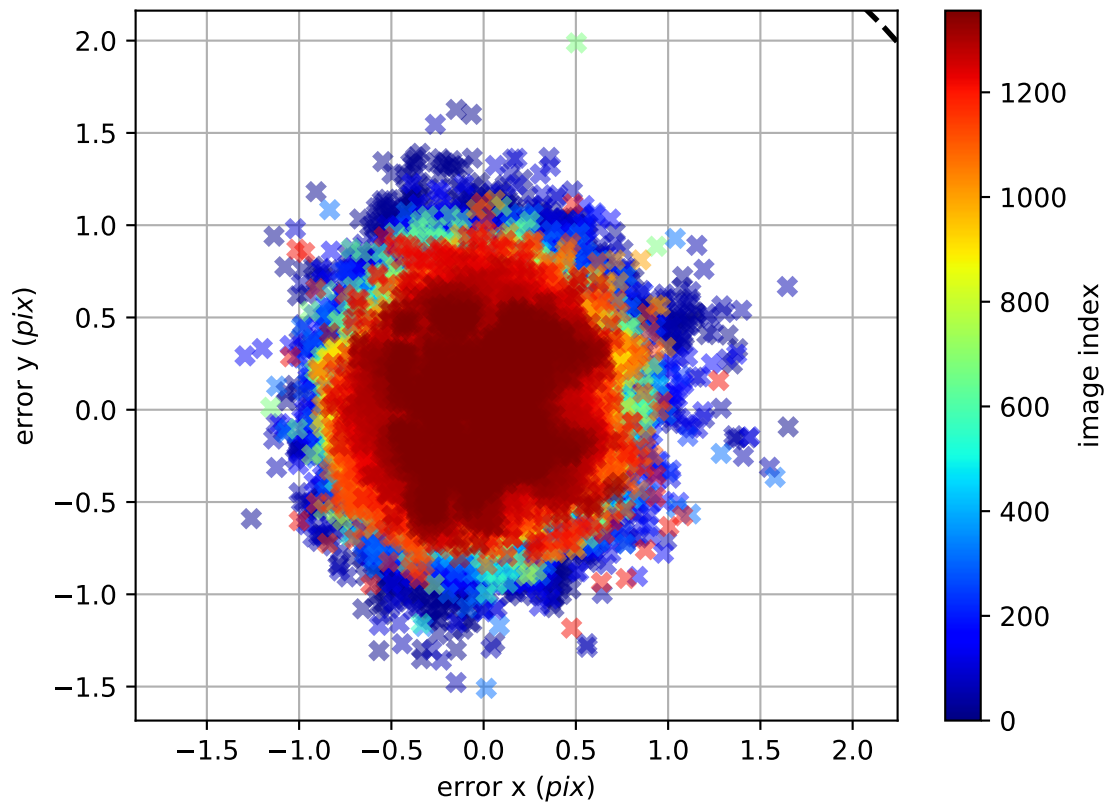
imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors



cam1: reprojection errors

